

AI Technologies in Government Sectors:

Introduction:

The use of AI in the U.S government is something that is relatively new. These technologies are responsible for some decision-making, national security management, and financial regulations for the entirety of the nation.

AI in the federal government is shifting from scattered experimentation to a coordinated, risk-managed, governance-driven enterprise. Agencies are adopting AI to improve efficiency, strengthen oversight, and enhance public services, while Congress and the Executive Branch are building a regulatory and organizational framework to ensure transparency, accountability, and safety.

Strategy:

This case study shows a clear evolution: AI is no longer treated as a niche technology but as a strategic capability requiring unified oversight. Executive Orders (especially EO 14110) establish expectations for responsible AI use. OMB memoranda (March & September 2024) operationalize those expectations by requiring risk assessments and

continuous monitoring, integrating AI governance into agency strategic and IT plans, mandating privacy officer involvement early in AI development, encouraging interagency collaboration and shared best practices, NIST's AI Risk Management Framework + GAI Profile provides technical and ethical standards for safe deployment. The federal government is building a layered governance system, with policy at the top, technical standards in the middle, and agency-level implementation at the bottom.

Risk Management - Ethics, Safety, Public Trust:

This segment of the study emphasizes that federal adoption is not “AI first”—it is risk-first.

Fairness: Avoiding bias in decision-making systems.

Transparency: Public disclosure of AI use cases (AI.gov inventory).

Accountability: Human oversight, auditability, and clear governance roles.

Privacy: Early involvement of privacy officers and compliance with federal privacy laws.

Security: Monitoring for cybersecurity risks and incident reporting.

The creation of Chief AI Officers across agencies is a major structural change. These officers act as internal stewards of responsible AI use, ensuring alignment with federal standards.

Synthesis - The Future of Government AI:

The federal government is moving toward a mature, regulated, and mission-driven AI ecosystem characterized by centralized governance (OMB, NIST, Chief AI Officers), distributed innovation (agency-specific AI applications), continuous risk management (privacy, security, fairness), public transparency (AI inventories, governance charters), and legislative momentum (bipartisan bills shaping future requirements).

AI is no longer optional—it is becoming a core component of federal operations. But the government's approach is intentionally cautious, emphasizing safety, accountability, and public trust.

Recommendations:

In order for AI in government to work collectively and effectively, there must be some ground rules set by humans which limit the abilities of the AI system. The government could pass laws that directly limit the abilities and usage of AI in the decision-making process. They could also

impose restrictions on AI as a whole, meaning that even the public would not have access to certain features of newer AI technologies. At the same time, limiting the abilities of AI has its downsides. New features would only be available to a select few, and these select few people could abuse their power and maliciously exploit the AI system. In the end, the government should be able to balance the positives of implementing AI into their systems with the negative consequences they may bring.

Conclusion:

Taken together, the federal government's approach to AI reflects a deliberate shift toward structured, accountable, and mission-aligned modernization. What began as scattered experimentation has matured into a coordinated ecosystem grounded in governance, technical standards, and continuous risk oversight. Executive Orders, OMB guidance, and NIST frameworks collectively ensure that innovation is balanced with fairness, transparency, privacy, and security, while new roles like Chief AI Officers embed responsibility directly into agency operations. As AI becomes an essential component of federal workflows, policymakers face the challenge of setting boundaries that protect the public without stifling progress or

concentrating power. Ultimately, the future of government AI will depend on maintaining this equilibrium—advancing technological capability while safeguarding public trust, democratic values, and the ethical foundations of federal decision-making.

Sources:

<https://www.ncsl.org/technology-and-communication/artificial-intelligence-in-government-the-federal-and-state-landscape>